



QUANTITATIVE SKILLS

QUESTION BANK

MODULE-1 (Short questions)

- 1) Find the sum of first 20 natural numbers?
- 2) what is the value of $1^2 + 2^2 + \dots + 10^2$?
- 3) what is the unit digit of $167 \times 2183 \times 497 \times 839 \times 235 \times 111 \times 1039 \times 251 \times 563$
- 4) Find the remainder $\frac{232 \times 43 \times 2347 \times 4671 \times 6988}{23}$
- 5) The number of zeros at the end of $80!$
- 6) Find the value of $8 \times 88 \times 888$
- 7) Find the LCM of 16, 18, 24 and 30.
- 8) Find the least number which when divided by 8, 9, 12 and 15 leaves the same remainder 5 in each case
- 9) simplify $\frac{423 \times 423 - 325 \times 325}{423 - 325}$
- 10) Find Total number of factors $N=4200$

MODULE-1 (Long questions)

- 1) The traffic signals at three different road crossings change after every 30, 45, 50 and 100 seconds respectively. If they all change simultaneously at 11:50 am, then at what time will they again change simultaneously?
- 2) Three wheels can complete 25, 21 and 16 revolutions per minute respectively. There is a red spot on each wheel that touches the ground at time zero. After how much time, all these spots will simultaneously touch the ground again?
- 3) 7 is added to a certain number and the sum is multiplied by 5. The product is then divided by 3 and 4 is subtracted from the quotient. If the result comes to 16, then what is the original number?
- 4) There are some students to be seated in an auditorium. When they were seated in rows of 8, 9, 11, and 12, there always were 3 students left out to be seated. But



when they were seated in 15 rows, none were left. Find the minimum number of students more than 1000.

5) The girls start jogging from the same point around a circular track and each one completes one round in 24 seconds, 36 seconds and 48 seconds respectively. After how much time will they meet at one point?

6) The cost of 4 bags and 3 boxes is Rs. 555 and the cost of 3 bags and 4 boxes Rs. 460. What is the cost of bag?

7) There are certain number of benches in a classroom. If four students sit on each bench three benches remain unoccupied. If, however, three students sit on each bench then three students remain standing in the class. Find the number of students in the class?

8) In a group of 70 people, 37 like coffee, 52 like tea each person likes at least one of the two drinks. Find the number of people who like both coffee and tea.

9) One-third of the boys and one-half of the girls of a college participated in a social work project. If the number of participating students is 300 out of which 100 are boys, what is the number of students in the college?

10) The value of $\sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + \sqrt{225}}}}}$ =

MODULE-2 (Short questions)

1) The salary of a worker is first increased by 25%, then it is decreased by 12%. Then percentage change in his salary is

2) Find 4% of Rs 3125.

3) After spending 69% of her money, a lady has Rs 93. How much had she at first?

4) If the price of one kg of rice is increased by 25%, the increase is Rs 12. Find the new price of rice per kg.



- 5) If A's salary is 20% more than that of B, then how much per cent is B's salary less than that of A?
- 6) If the profit made on a packet of tea is Rs 4 and the cost price of the packet is Rs 20, then how much is the profit percentage?
- 7) Harish bought a second-hand typewriter for Rs 1200 and spent Rs 200 on its repairs. He sold it for Rs 1680. What was his profit or loss per cent?
- 8) A dishonest fruit vendor professes to sell his goods at cost price but he uses a weight of 900 g for the kg weight. Find his gain per cent.
- 9) 50 successive discounts of 2%, 4%, 6%, 8% and so on will in a single discount of?
- 10) Find the single equivalent discount corresponding to the successive discounts of 10% and 15%?

MODULE-2 (Long questions)

- 1) We have an article costing Rs. 10000 which is its marked price or tagged price. When we give discount of 10% and 20%, successively on this article, the selling price we get after this?
- 2) The population of a town is 12500. It increases by 10% during the first year. During the second year, it decreases by 15% and increased by 20% during the third year. What is the population after 3 years?
- 3) In an examination, 33% of the students failed in Maths, 24% failed in English and 17% failed in both. Find the percentage of students who passed in both the subjects.
- 4) Find a single equivalent increase, if a number is successively increased by 10%, 20% and 30%.
- 5) In an examination 35% of the students failed in Math, 25% of the students failed in English, 45% of the students failed in Hindi. If 10% of the students failed in Math and English, 20% of the students failed in English and Hindi, 30% of the students failed in Math and Hindi and 5% of the students failed in all the three subjects Math, English and Hindi, then find the percentage of students who passed in all three subjects.



- 6) A man buys 3 tables and 12 chairs for Rs 2400. He sells the tables at a profit of 20% and chairs at a profit of 10% and makes a total profit of Rs 300. At what price did he buy tables and chairs?
- 7) $\frac{2}{3}$ of a commodity is sold at 30% profit, $\frac{1}{4}$ is sold at 16% profit and the rest at 12% profit. If a total profit of Rs 75 is earned, then find the value of the commodity.
- 8) Monoka started a business with an initial investment of Rs.500000. In the first year, he incurred a loss of 4%. However, during the second year, he earned a profit of 5% which in the third-year rise to 10%. Calculate his net profit for the entire period of three years.?
- 9) Pure ghee cost Rs 100 per kg. After adulteration it with vegetable oil costing Rs.50 per kg, a shopkeeper sells the mixture at rate of Rs. 96 per kg, thereby making a profit of 20%. In what ratio dose he mix the two?
- 10) If the manufacturer gains 10%, The wholesale dealer 15% and the retailer 25%, then find the cost of production of a table, the retailer price of which is Rs.1265?

MODULE-3 (Short questions)

- 1). A sum of Rs 800 is lent for one year at the rate of 18% per annum. Find the interest
- 2). A sum of Rs 4000 is lent for 5 years at the rate of 15% per annum. Find the interest
- 3). Anita borrowed Rs 400 from her friend at the rate of 12% per annum for $2\frac{1}{2}$ years. Find the interest and the amount paid by her.
- 4). A borrowed Rs 5000 from B at simple interest. After 4 years, B received Rs 1000 more than the amount given to B on loan. The rate of interest was:
- 5). If 1 Re produces Rs 9 as interest, in 60 years at simple interest, the rate percent is
- 6). Find the ratio of CI to SI on a certain sum at 5% per annum for 2 years.
- 7). If the CI on a certain sum for 2 years at 4% be Rs 510, what would be the SI?
- 8). At what rate per cent compound interest, will Rs 400 amount to Rs 441 in 2 years?



- 9). Rs 800 at 5% per annum compound interest amount to Rs 882 in
- 10). On what principal will the compound interest for 3 years at 5 percent amount to Rs 63.05?

MODULE-3 (Long questions)

- 1). Aniket deposited two parts of a sum of Rs 25000 in different banks at the rates of 15% per annum and 18% per annum respectively. In one year, he got Rs 4050 as the total interest. What was the amount deposited at the rate of 18% per annum?
- 2). A certain sum of money amounts to Rs 1125 in 5 years and to Rs 1200 in 8 years. Find the sum
- 3). A sum was put at SI at a certain rate for 4 years. Had it been put at 5% lower rate, it would have fetched Rs 100 less. Find the sum
- 4). Arun borrowed a sum of money from Jayant at the rate of 8% per annum simple interest for the first four years, 10% per annum for the next 6 years and 12% per annum for the period beyond 10 years. If he pays a total of Rs 12160 as interest only at the end of 15 years, how much money did he borrow?
- 5). Parameswaran invested an amount of Rs 12000 at the simple interest rate of 10 per cent per annum and another amount at the simple interest rate of 20 per cent per annum. The total interest earned at the end of one year on the total amount invested became 14 per cent per annum. find the total amount invested.
- 6) Raviraj invested an amount of Rs 10,000 at compound interest rate of 10 per cent per annum for a period of three years. How much amount will Raviraj get after 3 years?
- 7). Find the amount of Rs 1000 in 1 year at 5 percent compound interest payable half-yearly.
- 8). Find the amount of Rs 6400 in 1 year 6 months at 5 per cent compound interest, interest being calculated half yearly.
- 9). What will be the compound interest acquired on a sum of Rs 12000 for 3 years at the rate of 10% per annum?



10). Find the difference between the compound interest and the simple interest for the sum Rs 2500 at 6% per annum for 2 years.

MODULE-4 (Short questions)

- 1) If $3A = 4B = 5C$. Find $A : B : C = ?$
- 2) If $A : B = 4 : 5$, $b : c = 2 : 5$, then find $A : B : C = ?$
- 3) If $A : B = 5 : 6$ and $b : c = 4 : 7$ and $c : d = 3 : 1$, then find $A : B : C : d = ?$
- 4) Two numbers are in the ratio 4:9 If the both numbers increase by 12 the ratio becomes 11: 21. The sum of the original numbers is?
- 5) Find a fourth proportional to the numbers 5, 8, 15.
6. A and B started a business by investing 22500 and 35000, respectively. Out of a total profit of 27600. B's share is
- 7) If the product of the ages of M and N is 486 years and the ratio of their ages is 3: 2, what will be the sum of their ages after four years?
- 8) If 60% of $A = \frac{3}{4}$ of B, then $A : B$ is
- 9) Express 50 paise : Rs. 4 in its Ratio form?
- 10). A and B started a business by investing 22500 and 35000, respectively. Out of a total profit of 27600. B's share is

MODULE-4 (Long questions)

- 1) A, B and C spend 80%,85% and 75% of their incomes respectively. If their saving are in the ratio 8 : 9 : 20 and the difference between the incomes of A and c is Rs 18000, then the income of B is?
- 2) Four vessels of equal sizes contain mixture of spirit and water. The concentration of spirit in 4 vessels is 60%, 70%, 75% and 80% respectively. If all the four mixtures are mixed, find in the resultant mixture the ratio of spirit to water.
- 3). Mr. Sharma, Mr. Gupta and Ms. Sinha invested ₹4000, ₹8000 and ₹6000 respectively, in a business. Mr. Sharma left after 6 months. If after 8 months, there was a gain of 34000, then what will be the share of Mr. Gupta?



- 4) The ratio of present ages of Sita and her son is 5: 2 respectively. Seven years hence, the ratio will be 2:1. What was the age of Sita when her son was born?
- 5) The present age of Suresh is twice his daughter's present age. Six years hence, the ratio between the ages of Suresh and his daughter will be respectively 23: 13. What is the present age of Suresh?
- 6) There are 420 coins consist of one-rupee coins, 50 paise coins and 25 paise coins. If the ratio of their values is 2 : 3 : 5 , then the number of one-rupee coins is?
- 7) Three equal jugs are filled with a mixture of milk and water. The proportion of milk and water in each glass is in ratio 1 : 2 , 2 : 3 and 3 : 4. The contents of the three jugs are emptied a single vessel. What is the proportion of milk and water in it?
- 8) A sum of Rs1290 is divided between A, B and C such that A's share is $1\frac{1}{2}$ times that of B and B' share is $1\frac{3}{4}$ times that of C. What is C's share ?
- 9) Shankar started a business with an investment of Rs 120,000. After three months, Aniket joined him with an investment of Rs 190,000. They earned a profit of Rs 17,50,000 after one year. What is Aniket's share in the profit?
- 10) Of the three numbers, second is twice the first and is also third. If the average of three numbers is 44, what is the largest number?

MODULE-5 (Short questions)

- 1) A train runs at the rate of 45 km an hour. What is its speed in meters per second?
- 2) A motor car takes 50 seconds to travel 500 meters. What is its speed in km per hour?
- 3) How many km per hour does a man walk who passes through a street 600 m long in 5 minutes?
- 4) A train covers a distance in 50 minutes at an average speed of 48 km/h. At what speed must it run to cover the same distance in 40 minutes?
- 5) How long will a train 130 m long travelling at 40 km an hour, take to pass a kilometer stone?



- 6) How long will a train 60 m long travelling at 40 km an hour, take to pass through a station whose platform is 90 m long?
- 7) A train passes a pole in 15 seconds and passes a platform 100 m long in 25 seconds. Find its length.
- 8) A man riding his bicycle covers 150 metres in 25 seconds. What is his speed in km per hour?
- 9) The speed of a boat when travelling downstream is 32 km/hr, whereas when travelling upstream it is 28 km/hr. what is the speed of the boat in still water?
- 10) A 100-m long train is running at the speed of 30km/hr. Find the time taken by it to pass a man standing near the railway line?

MODULE-5 (Long questions)

- 1) A man covers a certain distance between his house and office on scooter. Having an average speed of 60 km/hr. he is late by 20 min. However, with a speed of 80km / hr. he reaches his office 10 min earlier. Find the distance between his house and office.
- 2) Two men A and B walk from P to Q, a distance of 22 km, at 5 and 6 km an hour respectively. B reaches Q, returns immediately and meets A at R. Find the distance from P to R.
- 3) Without any stoppage a person travels a certain distance at an average speed of 40 km/hr., and with stop-pages he covers the same distance at an average speed of 30 km/hr. How many minutes per hour does he stop?
- 4) A policeman goes after a thief who has 100 meters' start. If the policeman runs a kilometer in 8 minutes, and the thief a kilometer in 10 minutes, how far will the thief have gone before he is overtaken?
- 5) Ram travels at the rate of 3 km/hr. and he reaches 15 minutes late. If he travels at the rate of 4km / hr. he reaches 15 minutes earlier. The distance Ram has to travel is
- 6) Ramesh travel by bus from city A to city B at an average speed of 44km/hr. Suresh travel by taxi from city A to city B at an average speed of 77km/hr and takes 6 hours lesser than time taken by Ramesh. What is the distance (in km) between the two cities?



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- 7) A motor car does a journey in 12 hr. the first half at 15 km/hr. and the second half at 30 km/hr. Find the distance.
- 8) A man takes 6 hrs. 30 min in walking to a certain place and riding back. He would have gained 2 hrs. 10 min by riding both ways. How long would he take to walk both ways?
- 9) A train running at 35 km per hour takes 18 seconds to pass a platform. Next, it takes 12 seconds to pass a man walking at the rate of 5 km/hr. in the same direction. Find the length of the train and that of the platform.
- 10) A boat running upstream takes 10 hours to cover a certain distance, while it takes 7 hours to cover the same distance running downstream. What is the ratio between the speed of the boat and that of the water current respectively?